Code No: 114AG **R13**

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B.Tech II Year II Semester Examinations, April - 2018
FORMAL LANGUAGES AND AUTOMATA THEORY
(Computer Science and Engineering)

Time: 3 Hours **Max. Marks: 75**

Note: This question paper contains two parts A and B.
Part A is compulsory which carries 25 marks. Answer all questions in Part A.
Part B consists of 5 Units. Answer any one full question from each unit. Each question carries 10 marks and may have a, b, c as sub questions.

PART- A

(25 Marks)

1.a) Define NFA with $\in$-moves. [2]
b) Design FA to accept set of all strings over {0,1} containing 3 consecutive ones. [3]
c) Give recursive definition for Regular expression. [2]
d) If L1={a^n/n≥1} and L2= {b^n/n≥1} then find L1.L2. [3]
e) State pumping Lemma. [2]
f) Perform leftmost derivation for the string **aaabbabbba** with respect to the Grammar **S → aB / bA; A →a/ aS/ bAA; B→b / bS / aBB** [3]
g) Define Turing Machine. [2]
h) Write about Multi tape- Turing Machine. [3]
i) Write about Modified PCP. [2]
j) Give an example of a Language that is not recursive but recursively enumerable. [3]

PART-B

(50 Marks)

2.a) Convert the following NFA with € moves to DFA shown in figure 1.

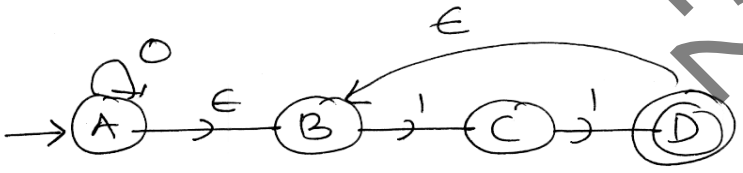


Figure: 1

b) Design Finite Automaton which accepts set of all strings not containing 101 as substring. [5+5]

OR

3.a) Design DFA for the language all strings over {0,1} which are ending with **00.**
b) Minimize the following DFA shown in figure 2. [4+6]

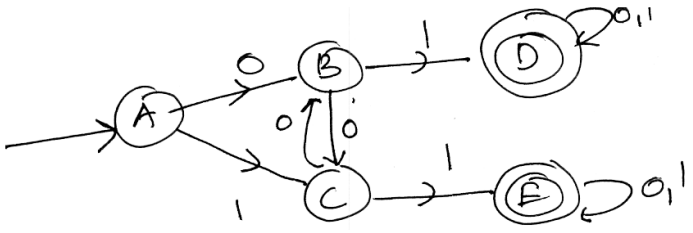


Figure: 2

JNTUH USED 26-04-2018PM

4.a) Convert the following Finite Automata to it's equivalent Regular Expression as shown in figure 3.

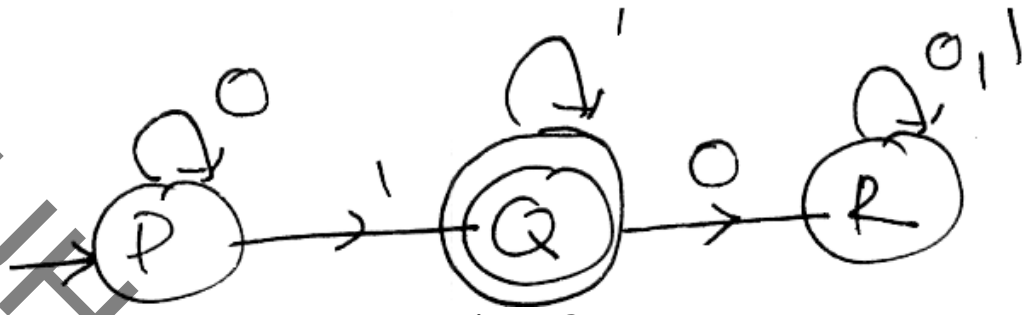


Figure: 3

b) Construct Finite Automata for the regular expression $\mathbf{0^*1^*(00)^*1}$. [5+5]

OR

5.a) Construct Finite Automata for ((0+1)*00)*110.

b) Find the right linear grammar for the following regular Expression **0*110*.** [5+5]

6.a) Design Push Down Automata for the language $\mathbf{L=\{wcw^R \mid w \in (0+1)^*\}}$

b) Convert the following Context Free Grammar (CFG) to Chomskey Normal Form (CNF)

S→AaB | aaB

A→€

B→bbA | € [5+5]

OR

7.a) Convert the following grammar to Chomsky Normal Form

S→aSa | bSb | a| b| aa | bb

b) Construct CFG for the PDA given below **A=({q0,q1},{0,1},{S,A},δ,q0,S,φ)** where **δ** is given as below [5+5]

δ (q0,1,S)={(q0,AS)}

δ (q0,ε,S)={(q0,ε)}

δ (q0,1,A)={(q0,AA)}

δ (q0,0,A)={(q1,A)}

δ (q0,1,A)={(q1,ε)}

δ (q1,0,S)={(q0,S)}

8.a) Construct the Turing machine for the language $\mathbf{L=\{ a^n b^{2n} \mid n>=1\}}$.

b) Write properties of recursively enumerable languages. [5+5]

OR

9.a) Explain about various types of Turing Machine.

b) Design Turing Machine for the Language L={$\mathbf{a^n\, b^n\, c^n / n \geq 1}$}. [5+5]

10.a) Write and explain about e Chomsky hierarchy of languages.

b) What do you mean by Halting Problem of Turing Machine? [7+3]

OR

11. Define the following and give examples

a) P – type problem

b) NP – Hard problem. [5+5]

---ooOoo---

JNTUH USED 26-04-2018PM